

Multi-source urban signals

311

Weather

POI

Land use

multi-source context

Leakage-safe panel

NTA x Week x Category
History at t

-> **Event at $t+1$**

shifted temporal features

$t \rightarrow t+1$ design

Prospective evaluation

LightGBM + XGBoost

2015-22 2023 2024-25



Train

Val.

Test

Category calibrated alerts

Ensemble score
-> service threshold

risk alerts

+ SHAP review

validation only